

HARDWARE TROUBLESHOOTING

Student Performance Analysis System

1. Introduction

A Student Performance Analysis System is a computer-based system used to collect, store, process, and analyze student academic information. It helps teachers and administrators monitor marks, attendance, grades, and overall student performance.

Since the system depends on various hardware components such as computers, keyboards, monitors, storage devices, network devices, and printers, hardware problems can affect the system's operation. Hardware troubleshooting is the process of identifying, diagnosing, and resolving these problems.

2. Hardware Components Used

The major hardware components required for a Student Performance Analysis System are:

- 1. Computer/Server – Runs the application and stores student information.
- 2. Monitor – Displays student records, reports, graphs, and analysis results.
- 3. Keyboard – Used to enter student details and marks.
- 4. Mouse – Used for navigation and selecting options.
- 5. Hard Disk/SSD – Stores the system software and student database.
- 6. RAM – Provides temporary memory required for processing data.
- 7. Network Router/Switch – Provides communication between computers in a network.
- 8. Printer – Used to print student performance reports.
- 9. UPS – Provides backup power and protects the system from sudden power failure.
- 10. Network/Internet Connection – Enables communication with online databases or cloud services when required.

3. Common Hardware Problems

S.No	Hardware Problem	Possible Cause	Solution
1	Computer does not start	Power failure or loose cable	Check power supply and cables
2	Monitor shows no display	Loose display cable or faulty monitor	Reconnect cable and test monitor

S.No	Hardware Problem	Possible Cause	Solution
3	Keyboard not working	Loose connection or damaged keyboard	Reconnect or replace keyboard
4	Mouse not responding	Connection or driver problem	Reconnect or replace mouse
5	System is very slow	Insufficient RAM or storage	Free storage or upgrade RAM
6	Database cannot be accessed	Storage or network problem	Check storage and network connection
7	Printer not printing	No paper, connection problem, or printer error	Check paper, cables, and printer status
8	Internet/network unavailable	Router or network cable problem	Restart router and check cables
9	System shuts down suddenly	Overheating or power problem	Check cooling system and UPS
10	Data cannot be saved	Insufficient storage or disk failure	Free storage or replace faulty drive

4. Hardware Troubleshooting Procedures

4.1 Computer Does Not Start

Problem

The computer does not power on when the power button is pressed.

Possible Causes

- o Power cable is disconnected.
- o Power socket is not working.
- o UPS is switched off.
- o Power supply unit is faulty.

Troubleshooting Steps

15. Check whether the power cable is properly connected.
16. Check the electrical socket.
17. Ensure that the UPS is switched on.
18. Check the power indicator.
19. Try another power cable if available.

20. If the system still does not start, contact a qualified technician.

4.2 Monitor Has No Display

Problem

The computer is running, but nothing appears on the monitor.

Possible Causes

- o Loose HDMI/VGA/DisplayPort cable.
- o Monitor power is disconnected.
- o Incorrect input source.
- o RAM or graphics hardware problem.

Troubleshooting Steps

25. Check the monitor power connection.
 26. Check the display cable at both ends.
 27. Select the correct input source on the monitor.
 28. Restart the computer.
 29. If there is still no display, test with another monitor or cable.
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4.3 Keyboard Not Working

Problem

The keyboard does not respond while entering student information or marks.

Possible Causes

- o Loose USB connection.
- o Damaged keyboard.
- o Faulty USB port.

Troubleshooting Steps

33. Disconnect and reconnect the keyboard.
 34. Try another USB port.
 35. Test the keyboard on another computer.
 36. Replace the keyboard if it is defective.
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4.4 Mouse Not Working

Problem

The mouse pointer does not move or respond.

Possible Causes

- o Loose USB connection.
- o Damaged mouse.
- o Faulty USB port.
- o Wireless mouse battery is low.

Troubleshooting Steps

41. Reconnect the mouse.
 42. Try another USB port.
 43. Replace the batteries if it is wireless.
 44. Test with another mouse.
 45. Replace the mouse if necessary.
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5. Slow System Performance

Problem

The Student Performance Analysis System takes a long time to open applications, load student records, or generate reports.

Possible Causes

- o Insufficient RAM.
- o Low free storage space.
- o Too many applications running.
- o Hard disk/SSD problems.
- o Overheating.

Troubleshooting Steps

51. Close unnecessary applications.
52. Delete unwanted files.
53. Maintain sufficient free disk space.
54. Check the system's RAM usage.
55. Upgrade RAM if required.
56. Check the health of the storage device.

57. Clean dust from cooling vents.

6. Network Connectivity Problem

Problem

The system cannot connect to the database server or other computers.

Possible Causes

- o Network cable disconnected.
- o Router/switch malfunction.
- o Wi-Fi problem.
- o Network configuration problem.

Troubleshooting Steps

- 62. Check the network cable.
 - 63. Check the router and switch indicators.
 - 64. Restart the router if necessary.
 - 65. Check the computer's network connection.
 - 66. Test another computer on the same network.
 - 67. Contact the network administrator if the problem continues.
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7. Printer Troubleshooting

Problem

Student performance reports cannot be printed.

Possible Causes

- o Printer is switched off.
- o No paper.
- o Empty ink/toner.
- o USB/network connection problem.
- o Paper jam.

Troubleshooting Steps

- 73. Turn on the printer.
- 74. Check the paper tray.
- 75. Check ink or toner levels.

- 76. Check the USB or network connection.
 - 77. Remove any paper jam carefully.
 - 78. Send the print job again.
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8. Storage Device Problems

Problem

Student records cannot be saved or retrieved.

Possible Causes

- o Insufficient disk space.
- o Hard disk/SSD failure.
- o File-system errors.
- o Storage device disconnected.

Troubleshooting Steps

- 83. Check available storage space.
- 84. Remove unnecessary files.
- 85. Check whether the storage device is detected.
- 86. Back up important student data.
- 87. Replace the storage device if it is failing.

Note: Important student records should always be backed up before attempting hardware repairs.

9. Overheating Problem

Problem

The computer becomes very hot, slows down, or shuts down unexpectedly.

Possible Causes

- o Dust accumulation.
- o Blocked ventilation.
- o Cooling fan failure.
- o High processor workload.

Troubleshooting Steps

- 92. Switch off the computer and allow it to cool.
- 93. Clean dust from ventilation areas.

- 94. Check whether cooling fans are operang.
- 95. Keep the computer in a well-venlated area.
- 96. Avoid blocking air vents.
- 97. Get professional assistance if the fan or cooling system is faulty.

10. UPS and Power Problems

Problem

The system shuts down during power interrupons.

Possible Causes

- o UPS baery is weak.
- o UPS is overloaded.
- o UPS is not connected properly.
- o Power supply problem.

Troubleshoo ng Steps

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|-----|--|
| 102 | Check the UPS power indicator. |
| . | Check the UPS baery condion. |
| 103 | Ensure that the computer is properly connected to the UPS. |
| . | Avoid connecng unnecessary high-power devices. |
| 104 | Replace the UPS baery when required. |

11. Hardware Troubeshoong Flow

Problem Detected

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Idenfy the Hardware Component

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Check Power and Connecons

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Check Cables and Ports

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Restart/Test the Device

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Check for Hardware Failure

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Repair or Replace the Component

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Test the Student Performance System

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Problem Resolved

12. Preventive Maintenance

Preventive maintenance helps reduce hardware failures and improves the reliability of the Student Performance Analysis System.

Recommended Practices

- Clean computers and peripherals regularly.
 - Keep ventilation areas free from dust.
 - Check cables and connections periodically.
 - Maintain sufficient free storage space.
 - Use a UPS to protect against power interruptions.
 - Keep backup copies of important student data.
 - Monitor system temperature.
 - Replace damaged hardware immediately.
 - Avoid installing unnecessary hardware devices.
 - Perform regular hardware inspections.
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13. Safety Precautions

While troubleshooting hardware, the following safety precautions should be followed:

- 117. Switch off and unplug the computer before opening its case.
- 118. Do not touch internal components unnecessarily.
- 119. Handle RAM, SSDs, and other components carefully.
- 120. Keep liquids away from computer equipment.
- 121. Use proper tools when performing hardware maintenance.
- 122. Do not attempt electrical repairs without proper knowledge.

- 123 Keep important student data backed up.
 - . Contact qualified technical personnel for serious hardware failures.
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14. Importance of Hardware Troubleshooting

Effective hardware troubleshooting is important for the Student Performance Analysis System because it:

- Reduces system downtime.
 - Improves system reliability.
 - Prevents data loss.
 - Ensures smooth data entry.
 - Helps generate reports without interruption.
 - Improves the performance of the system.
 - Reduces maintenance costs.
 - Provides a better experience for teachers and administrators.
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15. Conclusion

Hardware troubleshooting is an essential part of maintaining a Student Performance Analysis System. Problems involving computers, monitors, keyboards, storage devices, printers, networks, and power supplies can affect the availability and reliability of student information.

By identifying hardware problems early and following proper troubleshooting procedures, system downtime can be reduced and important academic data can be protected. Regular preventive maintenance, appropriate backups, and proper safety practices ensure that the Student Performance Analysis System operates efficiently and reliably.